

Definition : Let (Y, X) be a Random Vector. The Conditional Expectation of Y given $X = x$ is called Regression of Y on X .

$$\hat{y} = g(x, \beta) = E(Y|X=x)$$

Always Stochastic or Random Variable $\leftarrow \hat{y}$

Can be line, curve, surface, plane etc. $\leftarrow g(x, \beta)$

Can be Stochastic or Non-Stochastic $\leftarrow E(Y|X=x)$

$(k \times 1) =$

Linear & Non Linear Regression

Consider a data set $D = \{(x_i, y_i) \mid x_i \in \mathbb{R}^k, y_i \in \mathbb{R}, \forall i = 1, 2, \dots, n\}$ where all the x_i 's are non-stochastic, but y_i are stochastic & realized values of Random Variable Y_i 's respectively.

Definition :- If the relation, $g(x, \beta)$, between the response variable y & the regressor variable x is Linear in Parameter (β) , then it is called Linear Regression.

Eg:-

$$y = \beta_0 + \beta_1 x + \epsilon$$

Non-Linear Regression -

$$y = \beta_0 + \beta_1 e^x + \epsilon$$

$$\text{Eg:- } y = \frac{1}{\beta_0 + \beta_1 x} + \epsilon ; \quad y = \beta_0 \cos(\beta_1 + \beta_2 x) + \epsilon$$

Here, ϵ is Random error

Varieties of Regression : I.) Linear Model (LM)

- 1.) Simple Linear Regression
- 3.) Polynomial Regression

- 2.) Multiple Linear Regression

II.) Generalized Linear Model (GLM)

- 1.) Logit / Logistic Model
- 3.) Poisson - Regression

- 2.) Probit Model

Others :- Isotonic Regression, Spline Regression etc.

Simple Linear Regression

- Consider a dataset $D = \{(x_i, y_i) \mid x_i \in \mathbb{R}^k, y_i \in \mathbb{R}, \forall i = 1, 2, \dots, n\}$
- x_i 's are non stochastic
- y_i are stochastic & Realised values of Random Variable Y_i 's

Problem Statement : We are interested to have a Prediction line which will approximate well $\hat{y} = g(x, \beta_0, \beta_1) = \beta_0 + \beta_1 x$ the y values if the x values are known.

Minimum distance between the Predicted ($\hat{y}$'s) and the true (y 's) values of Y .

We will consider either Absolute or Square / Euclidean distance.

We will consider values of β_0 & β_1 that will minimise the Distance between the Predicted ($\hat{y}$'s) & the true (y 's) values of Y .

Least Squared Condition : The least squared condition to estimate the model parameters is to minimize $\rightarrow S(\beta_0, \beta_1) = \sum_i (y_i - \beta_0 - \beta_1 x_i)^2$ with respect to β_0 & β_1

If $(\hat{\beta}_0, \hat{\beta}_1)$ minimizes $S(\beta_0, \beta_1)$, then their values can be obtained by solving the Normal Equations :- $\frac{\partial S}{\partial \beta_0} = 0$ ① & $\frac{\partial S}{\partial \beta_1} = 0$ ②

Solving ① $\rightarrow -2 \sum_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$

Solving ② $\rightarrow -2 \sum_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) x_i = 0$

$$\Rightarrow \begin{aligned} \sum y_i &= n \hat{\beta}_0 + \hat{\beta}_1 \sum x_i \\ \sum x_i y_i &= \hat{\beta}_0 \sum x_i + \hat{\beta}_1 \sum x_i^2 \end{aligned}$$

On solving these we find :- $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$, Here :- $\bar{y} = \frac{1}{n} \sum y_i$ & $\bar{x} = \frac{1}{n} \sum x_i$

$$\& \hat{\beta}_1 = \frac{\sum x_i y_i - n \bar{x} \bar{y}}{\sum x_i^2 - n \bar{x}^2} = \frac{\sum y_i (x_i - \bar{x})}{\sum (x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}}$$

$\therefore$ The Prediction line will be $\rightarrow \hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$

The difference between the observed value y_i & the corresponding fitted value $\hat{y}_i$ is a Residual.

$$e_i = y_i - \hat{y}_i$$

Vector Notation

- Construct a data set $D = \{(x_i, y_i) \mid x_i \in \mathbb{R}, y_i \in \mathbb{R}\}$
- x_i 's are Non stochastic
- y_i 's are stochastic

$$y = (y_1, y_2, \dots, y_n)', \quad x = (x_1, x_2, \dots, x_n)', \quad \beta = (\beta_0, \beta_1)'$$

$$1 = (1, 1, \dots, 1)'$$

Problem Statement: We are interested to have a prediction vector:

It becomes a problem in $\mathbb{R}^n$ now

$$\hat{y} = g(x, \beta) = [1 \ x] \beta$$

$$y = f(x) + \epsilon$$

$$\underline{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \quad \underline{\epsilon} = \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{pmatrix}, \quad X = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix} = \begin{bmatrix} 1 & x \\ \sim & \sim \end{bmatrix}$$

$$\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}$$

$\therefore$

$$\underline{y} = X \underline{\beta} + \underline{\epsilon}$$

if $\hat{y} = g(x_1, x_2, \beta)$ Then :-

$$\underline{\tilde{y}} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \quad \underline{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}, \quad \underline{\epsilon} \text{ same as before}$$

$$\& \quad X = \begin{pmatrix} 1 & x_{11} & x_{21} \\ 1 & x_{12} & x_{22} \\ \vdots & \vdots & \vdots \\ 1 & x_{1n} & x_{2n} \end{pmatrix} = \begin{bmatrix} 1 & x_1 & x_2 \\ \sim & \sim & \sim \end{bmatrix}$$

So,

$$\underline{y} = X \underline{\beta} + \underline{\epsilon}$$

$$\text{or } y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \epsilon_i$$

This is Multiple Linear Regression.

Polynomial Regression

$$\underline{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_s \end{pmatrix}$$

$$\underline{X} = \begin{bmatrix} 1 & x_1 & x_1^2 & x_1^3 & \dots \\ 1 & x_2 & x_2^2 & x_2^3 & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \\ 1 & x_n & x_n^2 & x_n^3 & \dots \end{bmatrix}$$

It is also a Linear Regression

$$Y = X\beta + \epsilon$$

$$\text{or } y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{1i} x_{2i} + \beta_4 x_{1i}^2 + \beta_5 x_{2i}^2 + \epsilon_i$$

All of the above are Linear Models.

Vector Space $(V, +, \cdot)$

A vector space V over real numbers $\mathbb{R}$ is a collection of vectors such that

- 1.) $+$: $V \times V \rightarrow V$ (Closed under vector addition)
- 2.) $(x+y) + z = x + (y+z)$ for all $x, y, z \in V$ (Associative)
- 3.) There exists $0 \in V$ such that $\forall x \in V$ $0 + x = x + 0 = x$ [Identity element exists]
- 4.) There exists $-x \in V$ for each x such that $(-x) + x = x + (-x) = 0$ [Inverse exists]
- 5.) $x + y = y + x$ [Commutative]
- 6.) $a \cdot (b \cdot x) = (a \cdot b) \cdot x$ $\forall a, b \in \mathbb{R}$ & $x \in V$
- 7.) $1 \cdot x = x \cdot 1 = x$ $\forall x \in V$ [Multiplicative Identity]
- 8.) $(a+b) \cdot x = (a \cdot x) + (b \cdot x)$ $\forall a, b \in \mathbb{R}$ & $x \in V$
- 9.) $a \cdot (x+y) = a \cdot x + a \cdot y$

Subspace $(\mathcal{L}, +, \cdot)$

If a subset $\mathcal{L}$ of V is a vector space itself then $\mathcal{L}$ is called subspace of V .

- How to check $\mathcal{L}$ is a subspace of V ?
- 1.) Whether $0 \in \mathcal{L}$
 - 2.) Whether $x + a \cdot y \in \mathcal{L}$ $\forall x, y \in \mathcal{L}$ & $a \in \mathbb{R}$
- Example :-
- 1.) All lines passing through $(0,0)$ in $\mathbb{R}^2$
 - 2.) All planes passing through origin in $\mathbb{R}^n$
 - 3.) $\mathbb{P}_5$ in $\mathbb{P}_7$ — $\mathbb{P}_n$ is polynomial with degree upto n
- $\downarrow$
 $(\beta_0, \beta_1, \dots, \beta_5, 0, 0) \rightarrow (\beta_0, \beta_1, \beta_2, \dots, \beta_7)$

Span :- The span of a set vectors $\{v_1, v_2, \dots, v_k\} \in V$ is the collection :-

$$\text{Span}\{v_1, v_2, \dots, v_k\} = \left\{ \sum_{i=1}^k c_i v_i \mid c_i \in \mathbb{R} \right\}$$

which is the collection of all possible linear combinations of $v_1, v_2, \dots, v_k$

Note :- A span is always a subspace.

Example $\rightarrow$ a.) $\text{Sp}\{(0,1), (1,1)\} = \text{Sp}\{(0,1), (1,1)\} = \mathbb{R}^2$
 b.) $\text{Sp}\{(0,1,0), (1,0,0)\} = \mathbb{R} \times \mathbb{R} \times \{0\} = xy \text{ plane in } \mathbb{R}^3$

Linear Independence :- A set of vectors $\{v_1, v_2, \dots, v_n\} \in V$ are said to be linearly independent iff $\sum c_i v_i = 0 \Rightarrow c_1 = c_2 = \dots = c_n = 0$. On the other hand, if $\sum c_i v_i = 0$ holds for some non-zero $c_i \in \mathbb{R}$, the vectors are called Linearly dependent.

Example :- a.) $\{(0,1), (1,0)\}$ are Independent
 b.) $\{(0,1), (1,1)\}$ are Independent
 c.) $\{(0,1), (1,0), (1,1)\}$ are Dependent.

Basis :- If $\{v_1, v_2, \dots, v_k\}$ are linearly independent, then it is the basis of $\text{Sp}\{v_1, v_2, \dots, v_k\}$, and the dimension of $\text{Sp}\{v_1, v_2, \dots, v_k\}$ is the number of linearly independent elements in $\{v_1, v_2, \dots, v_k\}$.

Example :- a.) $\{(0,1), (1,1)\}$ is a basis of $\mathbb{R}^2$; Dimension = 2
 b.) $\{(0,1), (1,0)\}$ is a basis of $\mathbb{R}^2$ also. Dimension = 2
 c.) $\{(0,1), (1,0), (1,1)\}$ is NOT a basis of $\mathbb{R}^2$.

Orthogonal Vectors Two vectors $u, v \in V$ are said to be orthogonal if $u^T v = \sum u_i v_i = 0$

Orthogonal Complement :- If $\mathcal{L} \subseteq V$ is a subspace then the Orthogonal Complement of $\mathcal{L}$, denoted by $\mathcal{L}^\perp$ is a collection :-
 $\mathcal{L}^\perp = \{v \mid v \in V, u^T v = 0, \forall u \in \mathcal{L}\}$
 and $\dim(\mathcal{L}^\perp) = \dim(V) - \dim(\mathcal{L})$

Example :- a.) $\text{Sp}\{(1,0,0,0), (0,0,1,0)\} \perp \text{Sp}\{(0,1,0,0), (0,0,0,1)\}$
 b.) $\text{Sp}\{(1,1,0,0), (0,1,1,0), (1,0,1,0)\} \perp \text{Sp}\{(0,0,0,1)\}$

Remarks :- 1.) Basis is Not unique.
 2.) Elements of a basis need not be orthogonal to each other.
 3.) Linear Independence need not imply orthogonality.
 4.) Orthogonality implies independence.
 5.) Orthogonal vectors with unit length are called orthonormal vectors.

Projection

1.) If $\mathcal{L} \subseteq V$, then the Projection matrix of subspace $\mathcal{L}$ is $P_{\mathcal{L}}$ satisfying :-
 a.) $P_{\mathcal{L}} v = v$ if $v \in \mathcal{L}$

b) $P_S v \in S$ for all $v \in V$

2.) A Projection Matrix P_S is an Orthogonal Projection Matrix of subspace $S \subseteq V$ if $(I - P_S)$ is a projection matrix of $S^\perp \subseteq V$ too.

3.) If $\{v_1, v_2, \dots, v_k\}$ is an Orthonormal Basis of the subspace $S \subseteq V$ then the orthogonal projection matrix of S is $P_S = \sum v_i v_i^T$

Idempotency

1.) If a matrix P satisfies the relation that $P^2 = P$, then P is called an Idempotent Matrix.

2.) An Idempotent matrix has Eigenvalues 0 & 1.

3.) A Projection Matrix is an Idempotent Matrix

Note :- In regression, $\hat{y}$ eventually becomes the Orthogonal Projection of $y \in \mathbb{R}^n$ in the subspace $S = \text{span}\{1, x\}$

Column Space

The column space of a matrix $A = [a_1, a_2, \dots, a_n]$ with columns $a_1, a_2, \dots, a_n$ is :-

$$C(A) = \text{span}\{a_1, a_2, a_3, \dots, a_n\} = \{Ax \mid x \in \mathbb{R}^n\}$$

Hence, Row space of A denoted by $R(A) = C(A^T)$

Properties :- 1.) $C(A : B) = C(A) + C(B)$

2.) $C(AB) \subseteq C(A)$

3.) $\dim[C(A)] = \text{Rank}(A)$

4.) $C(AA^T) = C(A) \Rightarrow \text{Rank}(AA^T) = \text{Rank}(A)$

5.) If A has n -rows then $\dim(C(A)^T) = n - \text{Rank}(A)$

Quadratic Forms :-

A square symmetric matrix $A = (A_{ij})_{n \times n}$ is said to be -

- a.) Positive Definite (p.d.) if $x^T A x > 0$ for all $x \neq 0 \in \mathbb{R}^n$
 b.) Positive Semi-Definite (p.s.d.) if $x^T A x \geq 0$ for all $x \neq 0 \in \mathbb{R}^n$
 (Also called non-negative definite (n.n.d.))

Properties - a.) If A is p.d. then $|A| > 0$

b.) If A is p.s.d. then $|A| \geq 0$

Generalised Inverse :-

A matrix g is said to be a generalised inverse of a matrix A if $A g A = A$. Usually g is denoted by A^-

Properties - 1.) If A is $m \times n$ then A^{-1} is $n \times m$

2.) A^{-1} is not unique.

3.) For a matrix A , the Projection Matrix of $C(A)$ is AA^{-1}

4.) For a matrix A , the Orthogonal Projection Matrix of $C(A)$ is $A(A^T A)^{-1} A^T$.

Note - In regression, the prediction $\hat{y}$ and the error $y - \hat{y}$ are Orthogonal to each other.

Let $X = (X_1, X_2, \dots, X_n)^T$ be a Random Vector with finite expectation for each of the component, then we define Expectation of X as:-

$$E(X) = \begin{pmatrix} E(X_1) \\ E(X_2) \\ \vdots \\ E(X_n) \end{pmatrix}^T$$

We know :- $E(X) = \begin{cases} \int_{-\infty}^{\infty} x_i p(x_i) dx & \text{if } \sum_{i=1}^n \int_{-\infty}^{\infty} |x_i| p(x_i) dx < \infty \\ \int_{-\infty}^{\infty} x p(x) dx & \text{if } \int_{-\infty}^{\infty} |x| p(x) dx < \infty \end{cases}$

Similarly, if $Y = (Y_{ij})_{m \times n}$ is a random matrix with finite expectation for each of the component, then we define Expectation of a random matrix as:-

$$E(Y) = (E(Y_{ij}))_{m \times n}$$

$$\underline{X} = \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{pmatrix} \text{ then } E(\underline{X}) = \begin{pmatrix} E(X_1) \\ E(X_2) \\ \vdots \\ E(X_n) \end{pmatrix} \text{ if } E(X_i) \text{ exists for all } i = 1, \dots, n$$

Note - Component wise Median is not the median of the random vector

Dispersion Matrix :- It is also called Variance - Covariance Matrix

$$\begin{aligned} D(X) &= (Cov(X_i, X_j))_{n \times n} = Cov(X, X) = E[(X - E(X))(X - E(X))^T] \\ \text{We know } Var(X) &= E(X^2) - (E(X))^2 = E[(X - E(X))^2] \\ Cov(X, Y) &= E(XY) - E(X)E(Y) \\ &= E[(X - E(X))(Y - E(Y))] = Cov(Y, X) \\ \therefore Cov(X, X) &= Var(X) \end{aligned}$$

Note:- 1.) $Cov(U_p, V_q) = Cov((U_i, V_j))_{p \times q}$

$$2.) E(X+b) = E(X) + b$$

$$3.) D(X+b) = D(X)$$

$$4.) Cov(X+b, Y+c) = Cov(X, Y)$$

$$\begin{aligned} \text{Let } E(\underline{X}) &= \underline{\mu} ; \quad D(\underline{X}) = E[(\underline{X} - E(\underline{X}))(\underline{X} - E(\underline{X}))^T] \\ &= E[(\underline{X} - \underline{\mu})(\underline{X} - \underline{\mu})^T] \\ &= E(\underline{X} \underline{X}^T) - \underline{\mu} \underline{\mu}^T = (Cov(X_i, X_j))_{n \times n} \end{aligned}$$

- Note - 1.) Expectation is a location equivariant operation
 2.) Regression is a location invariant operation

Let X be a Random Vector with n - components such that $E(X) = \mu$ & $D(X) = \Sigma$

then :-
 1.) $E(L^T X) = L^T \mu$, where $L \in \mathbb{R}^n$ is a constant vector.

$$E(\underline{L}^T \underline{X}) = E\left(\sum_{i=1}^n l_i x_i\right) = \sum l_i E(x_i) = \underline{L}^T \underline{\mu}$$

$$2.) E(AX) = E(Y) = A \underline{\mu}; \quad A \text{ is } \mathbb{R}^{p \times n} \text{ constant matrix}$$

$$3.) D(L^T X) = L^T \Sigma L$$

$$4.) D(AX) = A \Sigma A^T \quad \& \quad \text{Cov}(AX, BX) = A \Sigma B^T$$

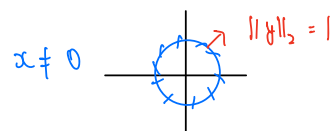
$$5.) \text{Cov}(V_p, V_q) = I, \text{ then } \text{Cov}(AV, BV) = A I B^T$$

Proof :- $D(\underline{L}^T \underline{X}) = \text{Cov}(\underline{L}^T \underline{X}, \underline{L}^T \underline{X}) = \text{Cov}(\sum l_i x_i, \sum l_i x_i)$
 $= \sum_{i=1}^n \sum_{j=1}^n l_i l_j \text{Cov}(x_i, x_j) = \underline{L}^T \Sigma \underline{L}$

Let $E(X) = \mu$, $D(X) = \Sigma$, It is easy to notice that $\Sigma = \Sigma^T$
 $D(X)$ will be a P. S. D. matrix.

Proof :- $Y = a^T X$; $X \in \mathbb{R}^n$ & $a \neq 0 \in \mathbb{R}^n$
 $\Rightarrow \text{Var}(Y) \geq 0$ $\therefore a^T \Sigma a \geq 0 \quad \forall a \neq 0$
 $\therefore \Sigma$ is a p.s.d.

Let $A = A^T$; $\Rightarrow \lambda_{\max} = \sup_{\|x\|_2=1} \frac{x^T A x}{x^T x}$
 $= \sup_{\|x\|_2 \leq 1} x^T A x$



Similarly, $\lambda_{\min} = \inf_{\|x\|_2=1} \frac{x^T A x}{x^T x} = \inf_{\|x\|_2 \leq 1} x^T A x$

Note :- $[0, 1] \rightarrow 0 = \min \quad \& \quad 1 = \max$
 $(0, 1) \rightarrow \min = \text{Undefined} \quad \& \quad 1 = \max$

Greatest Lower Bound $\Rightarrow \text{Inf}$

Similarly, Lowest Upper Bound $\Rightarrow \text{Sup}$; Inf & Sup need not belong to the set.



Theorem-1

Let X be a random vector with n -components such that $E(X) = \mu$ & $D(X) = \Sigma$. Then $\rightarrow$

$$E(X^T A X) = \text{tr}(\Sigma A) + \mu^T A \mu$$

Applications :- Case 1

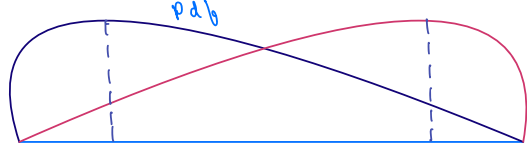
$$= E(\sum_{i=1}^n X_i^2) = \sum_{i=1}^n E(X_i^2) = \sum_{i=1}^n (\sigma_i^2 + \mu_i^2)$$

$\text{tr}(\Sigma I_n) \rightarrow \sum \sigma_i^2$ $\mu^T I_n \mu \rightarrow \sum \mu_i^2$

Case - 2 : $X_i \stackrel{iid}{\sim} N(0, 1) \Rightarrow \sum_{i=1}^n X_i^2 \sim \chi_n^2$; $A = I_n$, $\mu_i = 0 \forall i$
 It is clear that $\sum_{i=1}^n X_i^2 \sim \chi_n^2 \rightarrow$ Chi-Squared Distribution
 $E(\sum_{i=1}^n X_i^2) = n$

Now Consider $E(X) = \mu \neq 0 \Rightarrow E(\sum X_i^2) = n + \mu^T \mu > n$

Let $X_i \stackrel{iid}{\sim} N(0, 1)$; $Q_1 = X^T X = \sum_{i=1}^n X_i^2$. So $E(Q_1) = \sum_{i=1}^n E(X_i^2) = n$
 Let $Y_i \stackrel{iid}{\sim} N(\mu_i, 1)$; $Q_2 = Y^T Y = \sum_{i=1}^n Y_i^2$. So $E(Q_2) = n + \mu^T \mu$
 Degrees of Freedom Non Central Parameter



$E(Q_1) < E(Q_2)$ by a factor called Non-Central Parameter

Now, $E(X^T A X) = E(\text{tr}(X^T A X)) = E(\text{tr}(A X X^T)) = \text{tr}(A E(X X^T))$
 Real No. $= \text{tr}(A(\Sigma + \mu \mu^T)) = \text{tr}(A \Sigma) + \mu^T A \mu$

Theorem-2

Let X be a random vector with n -components such that $E(X) = \mu$ & $D(X) = \Sigma$. Then $\rightarrow$

$$P(X - \mu) \in C(\Sigma) = 1$$

In simple terms it tells us that Probability of Deviation / Fluctuation belonging to the Column Space of Variance-Covariance is 1.

Proof :- Let $l(A)$ be all possible linear combinations of the columns of A matrix.
 Also let $l \in [C(\Sigma)]^\perp \rightarrow$ Orthogonal Component of $C(\Sigma)$

$$\therefore l^T \Sigma = 0 \Rightarrow l^T \Sigma l = 0$$

$\therefore D(l^T (X - \mu)) = 0$; Now we know that $D(X + b) = D(X)$
 $\therefore D(l^T (X - \mu)) = 0 \Rightarrow \text{Var}(l^T (X - \mu)) = 0$ & $E(l^T (X - \mu)) = 0$
 This means for $X - \mu$, mean & var are both 0

So, $P(l^T (X - \mu) = 0) = 1$ OR $P[l \perp (X - \mu)] = 1$

$$P(X - \mu) \in C(\Sigma) = 1$$

Multivariate Normal

A random vector X is said to follow Multivariate Normal $N(\mu, \Sigma)$ if it has density:-
 p.d.f. $\sum$ $f(x) = \frac{\exp[-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)]}{(\sqrt{2\pi})^n \sqrt{|\Sigma|}}$ for some $\mu \in \mathbb{R}^n$ &

① If $X \sim N(\mu, \Sigma)$, then $AX \sim N(A\mu, A\Sigma A^T)$

Q.1.) Let $(X, Y) \sim$ Bivariate Normal $(\mu_x, \mu_y, \sigma_x^2, \sigma_y^2, \rho_{xy})$. Then obtain the p.d.f. $f(x, y)$

$$Z = \begin{pmatrix} x \\ y \end{pmatrix} \quad \mu = \begin{pmatrix} \mu_x \\ \mu_y \end{pmatrix} \quad ; \quad \Sigma = \begin{pmatrix} \sigma_x^2 & \rho \sigma_x \sigma_y \\ \rho \sigma_x \sigma_y & \sigma_y^2 \end{pmatrix}$$

$$\therefore f(Z) = \frac{\exp[-\frac{1}{2}(Z-\mu)^T \Sigma^{-1}(Z-\mu)]}{(\sqrt{2\pi})^2 \sqrt{|\Sigma|}}$$

$$Z-\mu = \begin{pmatrix} x-\mu_x \\ y-\mu_y \end{pmatrix} \quad ; \quad \Sigma^{-1} = \frac{1}{(1-\rho^2)\sigma_x^2\sigma_y^2} \begin{pmatrix} \sigma_y^2 & -\rho\sigma_y\sigma_x \\ -\rho\sigma_x\sigma_y & \sigma_x^2 \end{pmatrix}$$

$$(Z-\mu)^T = \begin{pmatrix} x-\mu_x & y-\mu_y \end{pmatrix}$$

$$\text{So, } f(x, y) = \frac{\exp\left[-\frac{1}{2}\left(\frac{1}{1-\rho^2}\right)\left[\left(\frac{x-\mu_x}{\sigma_x}\right)^2 + \left(\frac{y-\mu_y}{\sigma_y}\right)^2 - 2\rho\left(\frac{x-\mu_x}{\sigma_x}\right)\left(\frac{y-\mu_y}{\sigma_y}\right)\right]\right]}{2\pi (\sqrt{1-\rho^2}) \sigma_x \sigma_y}$$

We know that $\chi_n^2 = \text{gamma}\left(\frac{n}{2}, \frac{1}{2}\right)$; $E(f) = \frac{\alpha}{\lambda} = n$ & $V(f) = \frac{\alpha}{\lambda^2} = 2n$

$$f(x) = \frac{\lambda^\alpha e^{-\lambda x} x^{\alpha-1}}{\sqrt{\alpha}}$$

Theorem

If $X \sim N(\mu, I_n)$, then $X^T A X$ has Chi-Squared Distribution iff A is idempotent [$A^2 = A$]. Moreover, Degrees of Freedom = Rank(A) & Non Central Parameter = $\mu^T A \mu$

This theorem is called Cochran's Theorem

Corollary:- If A & A_1 are symmetric & idempotent matrices such that $A = A_1 + A_2$ & A_2 is a p.s.d matrix, then $X^T A_1 X$ & $X^T A_2 X$ are independently distributed

Example :- $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \sim N(0, I_3)$

$$A = I_3 = A_1 + A_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$X^T A X = X^T (A_1 + A_2) X = X^T A_1 X + X^T A_2 X$$

$$\Rightarrow \sum_{i=1}^3 x_i^2 = \sum_{i=1}^2 x_i^2 + x_3^2$$

$$\text{L.H.S} \Rightarrow \sum_{i=1}^3 x_i^2 \sim \chi^2_{df=3, nc=0}; \text{R.H.S} \Rightarrow X^T A_1 X \sim \chi^2_{df=2, nc=0} \quad X^T A_2 X \sim \chi^2_{df=1, nc=0}$$

$\therefore X^T A_1 X$ & $X^T A_2 X$ are Independently - distributed

Theorem

Let $X \sim N(\mu, I_n)$ & A is symmetric & $CA = 0$,
then $X^T A X$ & CX are independently distributed

T-Statistic

Let $x_i \stackrel{iid}{\sim} N(\mu, \sigma^2)$. Define :- $\bar{X} = \frac{1}{n} \sum_{i=1}^n x_i$
& $s^2 = \sum_{i=1}^n (x_i - \bar{X})^2$

① Find Distribution of $\bar{X}$ & s^2

$$X \sim N(\mu, \sigma^2 I_n); \quad \bar{X} = \frac{1}{n} \sum_{i=1}^n x_i^2 = \left(\frac{1}{n}\right) \mathbf{1}^T X$$

$$s^2 = \sum (x_i - \bar{X})^2 = X^T \left[I_n - \frac{1}{n} \mathbf{1} \mathbf{1}^T \right] X$$

It is easy to verify that : ① $s^2 = X^T A X$ ② $A^T = A$ ③ $A^2 = A$ ④ $\text{Rank}(A) = n-1$

Now, $CX \sim N(C\mu, C\Sigma C^T)$

$$C\mu = \frac{1}{n} \mathbf{1}^T \mathbf{1} \mu = \frac{1}{n} \mathbf{1}^T (\mu) \mathbf{1} = \mu$$

$$C\Sigma C^T = \frac{1}{n^2} \mathbf{1}^T \Sigma \mathbf{1} = \frac{1}{n^2} \mathbf{1}^T (\sigma^2 I_n) \mathbf{1} = \frac{\sigma^2}{n}$$

$$\text{Now, } X^T A X = \sigma^2 \left(\frac{X}{\sigma}\right)^T A \left(\frac{X}{\sigma}\right)$$

Here, we can see that $\rightarrow \frac{X}{\sigma} \sim N\left(\frac{\mu}{\sigma}, 1 \cdot I_n\right)$

$\therefore$ We can apply **Cochran's Theorem**

$$\sigma^2 \left[\left(\frac{X}{\sigma} \right)^T A \left(\frac{X}{\sigma} \right) \right] \sim \sigma^2 \chi^2 \begin{matrix} d.f = n-1 \\ n-c-p = \mu^T A \mu = 0 \end{matrix}$$

② Show that they are i.i.d. ; $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \Rightarrow \bar{X} - \mu \sim N\left(0, \frac{\sigma^2}{n}\right)$

$$\therefore \frac{\sqrt{n}}{\sigma} (\bar{X} - \mu) \sim N(0, 1)$$

Data :- It is a paired collection of $\{(y_i, x_i) \mid i=1, 2, \dots, n\}$

Model : $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ for $i=1, 2, \dots, n$

② $\varepsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$ Assumption

③ x_i 's are **Non stochastic**

Unknown :- β_0, β_1 & σ^2 have to be estimated

Matrix Representation :- $Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$; $\varepsilon = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}$; $\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}$

$$X_{n \times 2} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} ; \text{ Model: } Y = X \beta + \varepsilon$$

Here, $\varepsilon \sim N(0, \sigma^2 I_n)$ & (β, σ^2) are to be estimated

In general, $X = \begin{bmatrix} 1 & x_1 & \dots & x_m \end{bmatrix}$; $\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_m \end{pmatrix}$

$$\underset{(n \times 1)}{Y} = \underset{(n \times m)}{X} \underset{(m \times 1)}{\beta} + \underset{(n \times 1)}{\varepsilon} ; (Y - X\beta) \sim N(0, \sigma^2 I_n)$$

Least Squared Condition :- $Q(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2$

$$= (Y - X\beta)^T (Y - X\beta) = Y^T Y - Y^T X \beta - \beta^T X^T Y + \beta^T X^T X \beta$$

$$Q(\beta) = Y^T Y - 2 Y^T X \beta + \beta^T (X^T X) \beta$$

$$\frac{\partial Q}{\partial \beta} = 0 \Rightarrow X^T Y = (X^T X) \beta \rightarrow \text{Normal Eq.}$$

If exists

$$\therefore \hat{\beta}_{LS} = (X^T X)^{-1} X^T Y$$

Maximum Likelihood Estimation :- $f(y) = \frac{1}{(2\pi)^n \sqrt{|I_n|}} \exp\left(-\frac{1}{2} (Y - X\beta)^T I_n^{-1} (Y - X\beta)\right)$ → Part we care about

$$\ln(f(y)) = C - \frac{1}{2\sigma^2} (Y - X\beta)^T (Y - X\beta)$$

Under Normality $\rightarrow \hat{\beta}_{MLE} = \hat{\beta}_{LS}$

$$Y = X\beta + \epsilon \quad ; \quad \epsilon \sim N(0, \sigma^2 I_n)$$

$\Rightarrow$ Gauss Markov model : $Y \sim N(X\beta, \sigma^2 I_n)$ → Holds for all linear models

$$\hat{\beta}_{MLE} = \hat{\beta}_{LS} = \hat{\beta} = (X^T X)^{-1} (X^T Y)$$

Predicted Vector, $\hat{y}$ for the given data $X \rightarrow \hat{y} = X \hat{\beta} = [X(X^T X)^{-1} X^T] Y$
Orthogonal Projection Matrix for $C(X)$ ←

$$= P_X Y$$

Remarks :- 1.) We cannot observe the true error, i.e., ϵ

2.) But, we can find the estimated error as :- $\hat{\epsilon} = y - \hat{y} = y - P_X y$

$$= (I_n - P_X) y$$

3.) $P_X = P_X^T$ (Symmetric)

5.) $P_X (I_n - P_X) = 0$ Matrix

4.) $P_X = P_X^2$ (Idempotent)

6.) $\hat{\epsilon}^T \hat{y} = y^T (I_n - P_X)^T P_X y = 0$

Now, we know : $\hat{\beta} = (X^T X)^{-1} X^T Y$ where $Y \sim N(X\beta, \sigma^2 I_n)$
Non Stochastic

We want to find $\hat{\beta} \sim ?$
 Now, if $Z \sim N(\mu, \Sigma)$, then $AZ \sim N(A\mu, A\Sigma A^T)$

$$\therefore \hat{\beta} \sim N[(X^T X)^{-1} X^T \beta, (X^T X)^{-1} X^T \sigma^2 I_n [X^T X)^{-1} X^T]^T]$$

$$= (X^T X)^{-1} X^T X (X^T X)^{-1} X^T = (X^T X)^{-1}$$

So, $\hat{\beta} \sim N(\beta, \sigma^2 (X^T X)^{-1})$

Now, $\hat{\epsilon} = (I_n - P_X) y$ & $y \sim N(X\beta, \sigma^2 I_n)$

We want to find $\rightarrow \underline{e} \sim ?$

Can be seen that this is X

It can be seen that: $(I_n - P_x)X\beta = (X - \overset{\nearrow}{P_x X})\beta = 0$

Similarly, $(I_n - P_x)I_n \sigma^{-2} (I_n - P_x)^T = \sigma^{-2} (I_n - P_x)(I_n^T - P_x^T)$
 $= \sigma^{-2} (I_n - P_x)$

$\therefore \underline{e} \sim N(0, (I_n - P_x)\sigma^{-2})$

Now, $\hat{y} = X\hat{\beta} = P_x y$ [Predictor] & $\underline{e} = (I_n - P_x)y$ [Estimated Error]

$\therefore \text{Cov}(\hat{y}, \underline{e}) = \text{Cov}(P_x y, (I_n - P_x)y) \star$
 We know that $\text{Cov}(AX, BX) = A \Sigma B^T$ if $D(AX) = A \Sigma A^T$

So $\star$ will give $\rightarrow P_x \sigma^{-2} I_n (I_n - P_x)^T = 0$
 This means that $\hat{y}$ & $\underline{e}$ are Independently Distributed

As $E(\hat{\beta}) = \beta$; Hence $\hat{\beta}$ is an Unbiased Estimator of β

Estimator :- Function of Known Data only
 Set $z_1, z_2, \dots, z_n$ $\xrightarrow{\text{is a}} \theta(z) \rightarrow$ Parameter [Eg:- μ & σ^2 in case of Normal Distribution]

If an estimator $T(z)$ can be expressed as a linear combination of z with non-random coefficients, i.e., $T(z) = \sum_i \alpha_i z_i$, then $T(z)$ is called a Linear Estimator.
 Assuming $T(z_1, z_2, \dots, z_n)$ is an Estimator of θ , then if :-

$E(T(z_1, z_2, \dots, z_n)) = \theta \forall \theta$ in the parameter space, then $T(z_1, z_2, \dots, z_n)$ is called an Unbiased Estimator of θ .

Simple Linear Regression

Linear Estimator: If $T(y) = \sum \alpha_i y_i = \alpha^T y$, then $T(y)$ is a linear estimator

We want to show that $\rightarrow \hat{\beta}_1$ is a Linear Estimator

We know that $\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{\sum (y_i - \bar{y})(x_i - \bar{x})}{S_{xx}} = \frac{1}{S_{xx}} [\sum (x_i - \bar{x}) y_i] - \bar{y} \frac{(\sum (x_i - \bar{x}))}{S_{xx}}$

$\therefore \hat{\beta}_1 = \sum_{i=1}^n \alpha_i y_i \rightarrow$ Linear Estimator

$$\beta_1 = \alpha' y \quad ; \quad \alpha_i = \frac{(x_i - \bar{x})}{S_{xx}}$$

$$\hat{\beta}_1 \sim N(\alpha^T X \beta, \alpha^T (I_n - \sigma^2) \alpha) \quad ; \quad \text{Now } \alpha^T X \beta = \alpha^T \begin{bmatrix} 1 & x \end{bmatrix} \beta = \begin{bmatrix} \alpha^T 1 & \alpha^T x \end{bmatrix} \beta$$

Here, $\alpha^T 1 = \sum \alpha_i = 0$ & $\alpha^T x = \frac{1}{S_{xx}} \sum (x_i - \bar{x}) x_i = \frac{1}{S_{xx}} \sum (x_i - \bar{x})(x_i - \bar{x}) = 1$

$$\therefore \alpha^T X \beta = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \beta_1$$

Similarly, $\alpha^T (I_n - \sigma^2) \alpha = \sigma^2 \alpha^T \alpha = \sigma^2 \sum \alpha_i^2 = \sigma^2 \frac{1}{S_{xx}} \sum (x_i - \bar{x})^2 = \frac{\sigma^2}{S_{xx}}$

$$\therefore \hat{\beta}_1 \sim N(\beta_1, \frac{\sigma^2}{S_{xx}})$$

Note - 1) It can be shown that $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$
 2) $\hat{\beta}_0$ is a Linear & Unbiased Estimator.

$$3) \hat{\beta}_0 \sim N(\beta_0, \sigma^2 (\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}))$$

Prediction Line - $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x = \bar{y} - \hat{\beta}_1 \bar{x} + \hat{\beta}_1 x$

$$\therefore \hat{y} = \bar{y} + \hat{\beta}_1 (x - \bar{x}); \text{ Clearly Prediction Line passes through } (\bar{x}, \bar{y})$$

Estimation of σ^2

$$\varepsilon \sim N(0, \sigma^2 I_n) \rightarrow \text{Not observable}$$

$$\text{But } e = (y - \hat{y})^T (y - \hat{y}) = (I_n - P_X) y \sim N(0, (I_n - P_X) \sigma^2)$$

$$\text{Now, } \sum e_i^2 = (y - \hat{y})^T (y - \hat{y}) = (y - \hat{\beta} X)^T (y - \hat{\beta} X) = \sum (y_i - (\beta_0 + \beta_1 x_i))^2$$

Although ε_i 's are iid, e_i 's are dependent

$$\sum e_i^2 = e^T e = [(I_n - P_X) y]^T [(I_n - P_X) y] = y^T (I_n - P_X)^T (I_n - P_X) y$$

$$= y^T (I_n - P_X) y$$

$$\text{Now, we can write } e^T e = \sigma^2 \left(\frac{e}{\sigma} \right)^T \left(\frac{e}{\sigma} \right) = \sigma^2 \left(\frac{y}{\sigma} \right)^T (I_n - P_X) \left(\frac{y}{\sigma} \right)$$

By Cochran's Theorem, the above expression will have Chi-Squared Distribution

$$\frac{\underline{y}}{\sigma} \sim N\left(\frac{\underline{X}\beta}{\sigma}, I_n\right); \sigma^2 \left(\frac{\underline{y}}{\sigma}\right)' (I_n - P_X) \left(\frac{\underline{y}}{\sigma}\right) \sim \chi^2$$

D.O.F = Rank($I_n - P_X$)
n.c.p = $\left(\frac{\underline{X}\beta}{\sigma}\right)' (I_n - P_X) \left(\frac{\underline{X}\beta}{\sigma}\right)$

$$n.c.p = \beta' \frac{1}{\sigma} \underline{X}^T (I_n - P_X) \underline{X} \beta = \frac{\beta}{\sigma}' (\cancel{\underline{X}^T \underline{X}} - \cancel{\underline{X}^T \underline{X}}) \frac{\beta}{\sigma} = 0$$

$$D.O.F = \text{Rank}(I_n - P_X) = n-2$$

$$E(\underline{e}^T \underline{e}) = \sigma^2 (D.O.F + n.c.p.) = \sigma^2 (n-2) \Rightarrow E\left(\frac{\underline{e}^T \underline{e}}{n-2}\right) = \sigma^2$$

$$\text{Let } \hat{\sigma}^2 = \frac{\underline{e}^T \underline{e}}{n-2} \Rightarrow E(\hat{\sigma}^2) = \sigma^2$$

Thus, $\hat{\sigma}^2$ is an Unbiased Estimator of σ^2 .

$$\begin{aligned} \sum e_i^2 &= \sum (y_i - (\beta_0 + \beta_1 x_i))^2 = \sum_i (y_i - (\bar{y} - \hat{\beta}_1 \bar{x} - \hat{\beta}_1 x_i))^2 \\ &= \sum_i [(y_i - \bar{y}) - \hat{\beta}_1 (x_i - \bar{x})]^2 = S_{yy} + \hat{\beta}_1^2 S_{xx} - 2\hat{\beta}_1 S_{xy} \\ &= S_{yy} + \frac{S_{xy}^2}{S_{xx}} - 2 \frac{S_{xy} S_{xy}}{S_{xx}} \\ \therefore \sum e_i^2 &= S_{yy} - \frac{S_{xy}^2}{S_{xx}} \end{aligned}$$

$H_0: \beta_1 = b_1$ vs $H_1: \beta_1 \neq b_1$; b_1 is given in Q_2

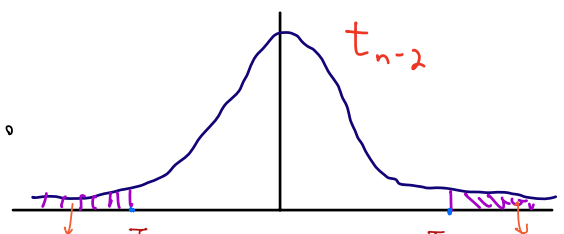
We know that $\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right) \Rightarrow \hat{\beta}_1 - \beta_1 \sim N\left(0, \frac{\sigma^2}{S_{xx}}\right)$

$$\therefore \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\frac{\sigma^2}{S_{xx}}}} \sim N(0, 1)$$

If H_0 / Null Hypothesis is true $\rightarrow \frac{\hat{\beta}_1 - b_1}{\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}} \sim N(0, 1)$ Under H_0

$$\text{We know :- } \hat{\sigma}^2 = \frac{\underline{e}^T \underline{e}}{n-2}$$

$$\text{So, } \frac{\hat{\beta}_1 - b_1}{\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}} \sim t_{n-2} \text{ under } H_0$$



We reject H_0 in favour of H_1 if:-

$$|T_{\text{observed}}| > t_{\frac{\alpha}{2}, n-2} \quad \text{at size } \alpha$$

$$\frac{\alpha}{2} = t_{\frac{\alpha}{2}, n-2}$$

$$t_{\frac{\alpha}{2}, n-2} = \frac{\alpha}{2}$$

$[\alpha \rightarrow \text{pre specified size of Test}]$

SLR

$$Y \sim N(X\beta, \sigma^2 I_n) ; \hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} \quad \& \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Now $y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$; for (x_0, y_0) : $y_0 \sim N(\beta_0 + \beta_1 x_0, \sigma^2)$
 Independent $\rightarrow$ Not i.i.d. due to x_i ; Unknown $\rightarrow$ Known

$\hat{y}_0$ is the Predicted value of y_0 given x_0 is known.
 $\hat{y}_0 = \hat{\beta}_1 x_0 + \hat{\beta}_0$

$$E(\hat{y}_0) = \hat{\beta}_0 + \hat{\beta}_1 x_0 \quad [\text{Not } y_0 \text{ as it itself is a R.V.}] = E(y_0)$$

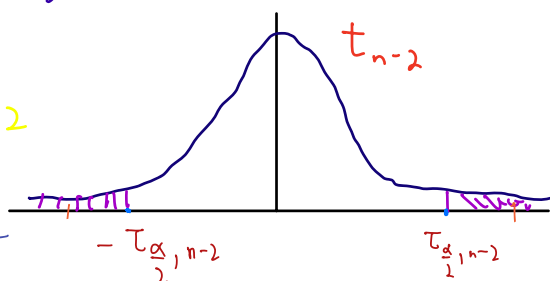
$$\begin{aligned} \text{Var}(\hat{y}_0) &= \text{Var}(\hat{\beta}_0 + \hat{\beta}_1 x_0) = \text{Var}(\bar{y} - \hat{\beta}_1 \bar{x} + \hat{\beta}_1 x_0) = \text{Var}(\bar{y} + \hat{\beta}_1 (x_0 - \bar{x})) \\ &= \text{Var}\left(\frac{1}{n} \sum_{i=1}^n y_i + (x_0 - \bar{x}) \frac{S_{xy}}{S_{xx}}\right) = \frac{1}{n^2} \times n \sigma^2 + \frac{(x_0 - \bar{x})^2}{S_{xx}} \times S_{xx} \sigma^2 \\ &\quad + 0 \rightarrow \text{Cov}(1^T, \alpha^T) = 0 \end{aligned}$$

y_0 is a random variable but $E(y_0)$ is a constant for given x_0 .

95% confidence interval of $E(y_0) = \beta_0 + \beta_1 x_0$

$$\hat{y}_0 \sim N(\beta_0 + \beta_1 x_0, \sigma^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right))$$

$$\Rightarrow \frac{\hat{y}_0 - (\beta_0 + \beta_1 x_0)}{\sqrt{\hat{\sigma}^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)}} \propto t_{n-2}$$



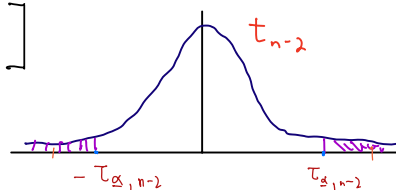
95% Confidence Interval of $(\beta_0 + \beta_1 x_0)$:-

$$\left(\hat{\beta}_0 + \hat{\beta}_1 x_0 - t_{\frac{\alpha}{2}, n-2} \sqrt{\hat{\sigma}^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)}, \hat{\beta}_0 + \hat{\beta}_1 x_0 + t_{\frac{\alpha}{2}, n-2} \sqrt{\hat{\sigma}^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)} \right)$$

Let y_0 & $(y_1, y_2, \dots, y_n)$ be Independent.

$$\Rightarrow y_0 - \hat{y}_0 \sim N \left[0, \sigma^2 \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right) \right]$$

$$\therefore T_0 = \frac{y_0 - \hat{y}_0}{\sqrt{\hat{\sigma}^2 \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)}} \sim t_{n-2}$$



$$P(-\tau_{\alpha/2, n-2} < t_0 < \tau_{\alpha/2, n-2}) = 1 - \alpha = 0.95$$

$$\therefore 95\% \text{ Prediction Interval of } y_0 \text{ is :- } \left[\hat{y}_0 \pm \tau_{\alpha/2, n-2} \sqrt{\sigma^2 \left(1 + \frac{1}{n} \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)} \right]$$

When Regressor is Stochastic

We know that :- $f(x, y) = \frac{1}{2\pi(1-\rho^2)} \exp \left[-\frac{1}{2(1-\rho^2)} \times Q(x, y) \right]$

Here $Q(x, y) = \left(\frac{x - \mu_x}{\sigma_x} \right)^2 + \left(\frac{y - \mu_y}{\sigma_y} \right)^2 - 2\rho \left(\frac{x - \mu_x}{\sigma_x} \right) \left(\frac{y - \mu_y}{\sigma_y} \right)$

Rearranging the terms $\rightarrow$

$$Q(x, y) = \left(\frac{y - \mu_y}{\sigma_y} \right)^2 - 2\rho \left(\frac{x - \mu_x}{\sigma_x} \right) \left(\frac{y - \mu_y}{\sigma_y} \right) + \rho^2 \left(\frac{x - \mu_x}{\sigma_x} \right)^2 + (1 - \rho^2) \left(\frac{x - \mu_x}{\sigma_x} \right)^2$$

$$= \left[\left(\frac{y - \mu_y}{\sigma_y} \right) - \rho \left(\frac{x - \mu_x}{\sigma_x} \right) \right]^2 + (1 - \rho^2) \left(\frac{x - \mu_x}{\sigma_x} \right)^2$$

$$\therefore Q(x, y) = \frac{1}{\sigma_y^2} \left[y - \mu_y - \rho \frac{\sigma_y}{\sigma_x} (x - \mu_x) \right]^2 + (1 - \rho^2) \left(\frac{x - \mu_x}{\sigma_x} \right)^2$$

Substituting $Q(x, y)$ in $f(x, y)$ we get :-

$$f(x, y) = \frac{\exp \left[-\frac{1}{2(1-\rho^2)} \times \left(\frac{1}{\sigma_y^2} \left[y - \mu_y - \rho \frac{\sigma_y}{\sigma_x} (x - \mu_x) \right]^2 + (1 - \rho^2) \left(\frac{x - \mu_x}{\sigma_x} \right)^2 \right) \right]}{\sqrt{2\pi} (\sigma_y \sqrt{1-\rho^2}) \sqrt{2\pi} \sigma_x}$$

$$= \frac{\exp \left[-\frac{1}{2(1-\rho^2)} \times \frac{1}{\sigma_y^2} \left[y - \left(\mu_y + \rho \frac{\sigma_y}{\sigma_x} (x - \mu_x) \right) \right]^2 \right]}{\frac{1}{\sqrt{2\pi}} \sqrt{\sigma_y^2 (1-\rho^2)}}$$

Conditional

$$\times \frac{\exp \left[-\frac{1}{2} \left(\frac{x - \mu_x}{\sigma_x} \right)^2 \right]}{\frac{1}{\sqrt{2\pi}} \sigma_x} \rightarrow \text{Marginal}$$

Hence $Y|X \sim N \left(\mu_y + \rho \frac{\sigma_y}{\sigma_x} (x - \mu_x), \sigma_y^2 (1 - \rho^2) \right)$

$\text{Var}(Y) = \sigma_y^2 \geq (1 - \rho^2) \sigma_y^2 = \text{Var}(Y|X)$

$$\begin{pmatrix} Y \\ X \end{pmatrix} \sim N \left[\begin{pmatrix} \mu_y \\ \mu_x \end{pmatrix}, \begin{pmatrix} \sum y y & \sum y x \\ \sum y x & \sum x x \end{pmatrix} \right], Y|X=x \sim N \left(\mu_y + \frac{\sum y x \sum x x (x - \mu_x)}{\sum y y - \sum y x \sum x x^{-1} \sum y x}, \frac{\sum y y - \sum y x \sum x x^{-1} \sum y x}{\sum x x} \right)$$

Now, If $(X, Y) \sim \text{BVN}(\mu_x, \mu_y, \sigma_x^2, \sigma_y^2, \rho)$ [To check Independence of X & Y]

① $H_0 : \rho = 0$ (Independent)

For Testing $H_1 : \rho \neq 0$

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}} = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}$$

(sample correlation)

$$T = \frac{r \sqrt{n-2}}{\sqrt{1-r^2}} \sim t_{n-2} \text{ Under } H_0$$

② $H_0 : \rho = \rho_0 \neq 0$
 $H_1 : \rho \neq \rho_0$

$Z = \ln\left(\frac{1+r}{1-r}\right)$ $\tanh^{-1}$ Transformation

$\mu_0 = \ln\left(\frac{1+\rho_0}{1-\rho_0}\right)$ & $\sigma_z^2 = (n-3)^{-1}$

$W = \frac{Z - \mu_0}{\sigma_z} \sim N(0, 1) \text{ as } n \rightarrow \infty$
 Asymptotic Test

Multiple Linear Regression

$Y \sim N(X\beta, \sigma^2 I_n)$; $Y \in \mathbb{R}^n$, $\varepsilon \in \mathbb{R}^n$

$X \in \mathbb{R}^{n \times (k+1)}$, $\beta \in \mathbb{R}^{k+1}$, $\sigma^2 > 0$

Hence, there are $k+2$ unknown parameters, β & $\sigma^2 > 0$
 ★ Similar to SLR, after we Minimise $\beta_{LS} = \hat{\beta}$

$\hat{\beta} = (X^T X)^{-1} X^T Y$; So $\hat{y} = X \hat{\beta} = X (X^T X)^{-1} X^T Y = P_X Y \rightarrow$ Projection of y in $C(X)$

$e = y - \hat{y} = (I_n - P_X) Y$; $e^T \hat{y} = 0$

$\hat{y} \sim N(X\beta, \sigma^2 P_X)$; $e \sim N(0, \sigma^2 (I_n - P_X))$
 Here $e \in C(X)^\perp = C(X^T X)^\perp$

Hence, $\hat{y}$ & e are uncorrelated & they are independently distributed when $\varepsilon \sim N(0, \sigma^2 I_n)$

Linear Unbiased Estimator

of $p^T \beta$

if

$$E(L^T y) = p^T \beta$$

for all

$$\beta \in \mathbb{R}^{k+1}$$

A linear estimator $L^T y = \sum l_i y_i$ is said to be a **LVE** if $E(L^T y) = p^T \beta$ for all $\beta \in \mathbb{R}^{k+1}$

Linear Zero Function

A linear estimator $L^T y = \sum l_i y_i$ is said to be a **LZF** if $E(L^T y) = 0$ for all $\beta \in \mathbb{R}^{k+1}$

BLUE

The Best Linear Unbiased Estimator (BLUE) of a linear parameter function $p^T \beta$ is a LVE with minimum variance

Theorem - A linear function is BLUE of its expectation iff it is uncorrelated with all linear zero function (LZF)

★ If $L^T y$ is an LVE of $p^T \beta$, then $L^T P_X y$ is the BLUE of $p^T \beta$

Analysis of Variance [ANOVA]

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_R = 0$$

$$H_1 : H_0 \text{ is not true}$$

→ Under H_0 , $y_i \stackrel{iid}{\sim} N(\beta_0, \sigma^2)$ we use $\bar{y}$ to estimate it.

$$SS_{\text{Total}} = \sum (y_i - \bar{y})^2 = Y^T \left(I_n - \frac{1}{n} \mathbf{1} \mathbf{1}^T \right) Y \quad [\text{Model building is not needed}]$$

$$SS_{\text{Error}} = \sum (y_i - \hat{y}_i)^2 = (y - \hat{y})^T (y - \hat{y}) = [(I_n - P_X) Y]^T (I_n - P_X) Y = Y^T (I_n - P_X) Y$$

$$SS_{\text{Model}} = \sum (\hat{y}_i - \bar{y})^2 = Y^T \left(P_X - \frac{1}{n} \mathbf{1} \mathbf{1}^T \right) Y = \frac{SS_{xy}^2}{SS_{xx}}$$

We can notice that : $SS_{\text{Total}} = SS_{\text{Error}} + SS_{\text{Model}}$

By Cochran's Theorem :-

$$\frac{SS_{\text{Error}}}{\sigma^2} \sim \chi^2_{d.f. = n-R-1} \quad [Rank(I_n - P_X) = n-R-1]$$

$$\frac{SS_{\text{Model}}}{\sigma^2} \sim \chi^2_{d.f. = b}$$

$$\frac{SS_{\text{Total}}}{\sigma^2} \sim \chi^2_{d.f. = n-1}$$

$$ncp = \lambda \geq 0$$

$$ncp = \lambda \geq 0$$

$$\text{Under } H_0 : \lambda = 0$$

$$H_1 : \lambda > 0$$

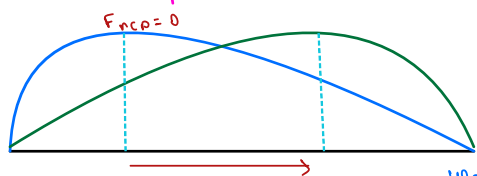
$\Rightarrow$

$$\frac{SS_{\text{Model}} / k}{SS_{\text{Error}} / (n-k-1)}$$

$$\sim F_{k, n-k-1 ; ncp=0}$$

Under H_0

$$F_{k, n-k-1 ; ncp=\lambda > 0} \text{ Under } H_1$$



We will reject H_0 in favour of H_1 if :-
 $F_{\text{obs}} > F_{\alpha, k, n-k-1}$

$$\star H_0 : \beta_j = b_j ; \hat{\beta}_j = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}^T \hat{\beta} \sim N(l_j^T \beta, \sigma^2 (l_j^T C l_j))$$

$$H_1 : \beta_j \neq b_j$$

$$C = (X^T X)^{-1}$$

$$\begin{pmatrix} 0 \\ \vdots \\ 1 \end{pmatrix} \rightarrow j^{\text{th}}$$

$$= N(\beta_j, \sigma^2 (C_{jj}))$$

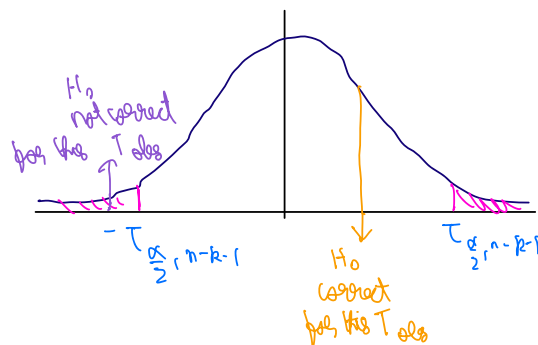
$$\text{Thus, } \hat{\beta}_j - \beta_j \sim N(0, \sigma^2 C_{jj}) \Rightarrow \frac{\hat{\beta}_j - \beta_j}{\sqrt{\sigma^2 C_{jj}}} \sim N(0, 1)$$

But σ^2 is Unknown.

$$\hat{\sigma}^2 = \frac{SS_{\text{Error}}}{n-k-1} = MSE_{\text{Error}} = \frac{Y^T (I_n - P_X) Y}{n-k-1}$$

$$\therefore \frac{\hat{\beta}_j - \beta_j}{\sqrt{\hat{\sigma}^2 C_{jj}}} = \frac{(\hat{\beta}_j - \beta_j) / \sqrt{\sigma^2 C_{jj}}}{\sqrt{\left(\frac{Y^T (I_n - P_X) Y}{\sigma^2} \right) / (n-k-1)}} \sim t_{n-k-1}$$

$$\text{Under } H_0 : \frac{\hat{\beta}_j - \beta_j}{\sqrt{\hat{\sigma}^2 C_{jj}}} \sim t_{n-k-1}$$



$$\star\star H_0 : \beta_i - \theta \beta_j = 0$$

$$H_1 : \beta_i - \theta \beta_j > 0$$

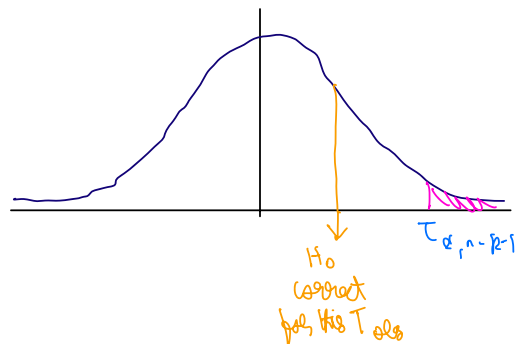
$$\beta_i - \theta \beta_j = \underline{l}^T \underline{\beta}$$

$$; \underline{l} = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ -\theta \\ \vdots \\ 0 \end{pmatrix} \rightarrow i \text{th} \rightarrow j \text{th}$$

$$\underline{\underline{L}}^T \hat{\underline{\underline{\beta}}} \sim N(\underline{\underline{L}}^T \underline{\underline{\beta}}, \sigma^2 \underline{\underline{L}}^T (\underline{\underline{L}})) = N(\beta_i - \theta \beta_j, \sigma^2 (c_{ii} - 2\theta c_{ij} + \theta^2 c_{jj}))$$

$$So, \frac{\underline{\underline{L}}^T \hat{\underline{\underline{\beta}}} - \underline{\underline{L}}^T \underline{\underline{\beta}}}{\sqrt{\hat{\sigma}^2 (c_{ii} - 2\theta c_{ij} + \theta^2 c_{jj})}} \sim t_{n-k-1}$$

Under H_0



We reject H_0 in favour of H_1 at level α iff :-

$$T_{obs} > T_{\alpha, n-k-1}$$

$$*** H_0 : \begin{cases} \beta_i = 0 \\ \beta_j = 0 \end{cases}$$

or

$$H_0 : \begin{cases} \beta_i = 0 \\ \beta_i - \beta_j = 0 \end{cases}$$

$$A \underline{\underline{\beta}} = 0 \rightarrow \begin{pmatrix} 0 & 0 & \dots & 1 & \dots & 0 \\ 0 & 0 & \dots & 1 & -1 & \dots & 0 \end{pmatrix}$$

$$B \underline{\underline{\beta}} = 0 \rightarrow \begin{pmatrix} 0 & 0 & \dots & 1 & \dots & 0 \\ 0 & 0 & \dots & 1 & -1 & \dots & 0 \end{pmatrix}$$

$$A \hat{\underline{\underline{\beta}}} \sim N(A \underline{\underline{\beta}}, \sigma^2 A (A^T)^{-1} A) \Rightarrow A \hat{\underline{\underline{\beta}}} - A \underline{\underline{\beta}} \sim N(0, \sigma^2 A (A^T)^{-1} A)$$

By Cochran's Theorem :- $(A \hat{\underline{\underline{\beta}}} - A \underline{\underline{\beta}})^T (\sigma^2 A (A^T)^{-1} A)^{-1} (A \hat{\underline{\underline{\beta}}} - A \underline{\underline{\beta}}) \sim \chi^2_{\text{Rank}(A)}$
 $n - p = 0$

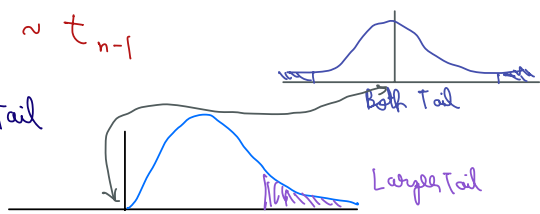
$$\therefore \frac{(A \hat{\underline{\underline{\beta}}} - A \underline{\underline{\beta}})^T (A (A^T)^{-1} A)^{-1} (A \hat{\underline{\underline{\beta}}} - A \underline{\underline{\beta}}) / \text{Rank}(A)}{(Y^T (I_n - P_X) Y) / (n-k-1)} \sim F_{dF_1 = \text{Rank}(A), dF_2 = n-k-1}$$

$$\text{Under } H_0 : \frac{(A \hat{\underline{\underline{\beta}}} - \underline{\underline{b}})^T (A (A^T)^{-1} A)^{-1} (A \hat{\underline{\underline{\beta}}} - \underline{\underline{b}}) / \text{Rank}(A)}{(Y^T (I_n - P_X) Y) / (n-k-1)} \sim F_{dF_1 = \text{Rank}(A), dF_2 = n-k-1}$$

$b = 0$ in our case; Now for testing : $X \sim N(\mu, \sigma^2)$
 $H_0 : \mu = \mu_0, H_1 : \mu \neq \mu_0$

$$\text{Note :- } T = \frac{\sqrt{n} (\bar{X} - \mu_0)}{\hat{\sigma}} \sim t_{n-1}$$

$T^2 \sim F_{1, n-1} \Rightarrow \text{Only upper Tail}$



★★★★

$$H_0: \beta_1 = \beta_2 = \dots = \beta_k = 0$$

$$H_1: H_0 \text{ is not true.}$$

$$A = \begin{bmatrix} 0 & 1 & \dots & 0 \\ 0 & 0 & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{bmatrix}_{k \times (k+1)} \quad X \beta = 0$$

Under H_0 :

$$\frac{(A\hat{\beta} - 0)^T (A(A^T A)^{-1} A^T \hat{\beta} - 0)}{(Y^T (I_n - P_X) Y) / (n-k-1)} \sim F_{\substack{dF_1 = k \\ dF_2 = n-k-1}}$$

We can also write the F statistic in another way.

$$F = \frac{\frac{SS_{\text{Model}}}{\sigma^2} / k}{\frac{SS_{\text{Error}}}{\sigma^2} / (n-k-1)} \quad ; \quad \begin{matrix} SS_{\text{Total}} & = & SS_{\text{Model}} & + & SS_{\text{Error}} \\ (n-1) & & (k) & & (n-k-1) \\ ncp(0) & & ncp(\lambda) & & ncp(0) \end{matrix}$$

Now, we know: $\frac{Y}{\sigma} \sim N\left(\frac{XB}{\sigma}, I_n\right)$

$$SS_{\text{Model}} = \sum (\hat{y}_i - \bar{y})^2 = Y^T \left(P_X - \frac{1}{n} \mathbf{1} \mathbf{1}^T \right) Y$$

$$\text{So, } ncp\left(\frac{SS_{\text{Model}}}{\sigma^2}\right) = \left(\frac{XB}{\sigma} \right)^T \left(P_X - \frac{1}{n} \mathbf{1} \mathbf{1}^T \right) \left(\frac{XB}{\sigma} \right) = \frac{1}{\sigma^2} \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} \begin{bmatrix} 0 & 0^T \\ 0 & X_R^T X_R - \frac{1}{n} X_R^T \mathbf{1} \mathbf{1}^T X_R \end{bmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}$$

Here, $\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}$ & $X = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} X_R$

$$\text{Now, } \frac{1}{(n-1)} \left[X_R^T X_R - \frac{1}{n} X_R^T \mathbf{1} \mathbf{1}^T X_R \right] = \text{Cov}(\underline{X})$$

For $ncp = 0 \Rightarrow \beta_R = 0$

Polynomial Linear Regression

Consider a linear model $\rightarrow y_i = \beta_0 + \beta_1 x_i^1 + \beta_2 x_i^2 + \dots + \beta_k x_i^k + \epsilon_i \quad \forall i=1, 2, \dots, n$
 We want to estimate β which will minimise least squared condition.

In Polynomial Linear Regression, we consider Higher degree of component x , but it is linear in parameters.

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \epsilon \quad \text{For single regressor}$$

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1^2 + \beta_4 x_2^2 + \beta_5 x_1 x_2 + \epsilon \quad \text{For Multiple Regressor}$$

Orthogonal Polynomial

to be Orthogonal Polynomial if:-

- For a given set of input data $x = (x_1, x_2, \dots, x_n)$, a set of polynomials $\{p_0, p_1, \dots, p_k\}$ are said

$$p_0(x_i) = 1 \quad \& \quad \sum p_j(x_i) p_k(x_i) = 0 \quad \forall j \neq k$$

So, our model is $\longrightarrow y_i = \sum_{j=0}^k \alpha_j p_j(x_i) + \epsilon_i \quad \forall i = 1, 2, \dots, n$

$\Rightarrow Y = Z\alpha + \epsilon$; When $Y \sim N(Z\alpha, \sigma^2 I_n)$, the least squares estimate will be:-

$$\hat{\alpha} = (Z^T Z)^{-1} Z^T Y$$

Coefficient of Determination

For a given set of data (y, x) , we first fit linear model under standard assumptions. To measure how good the model is to explain dependencies between regression & independent variables, we use Coefficient of Determination (R^2)

$$R^2 = \frac{\text{Variation of } Y \text{ explained by the model}}{\text{Total variation in } Y} = \frac{\sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2}{\sum_{i=1}^n (Y_i - \bar{Y})^2} = \frac{SS_{\text{Model}}}{SS_{\text{Total}}}$$

$$= 1 - \frac{SS_{\text{Error}}}{SS_{\text{Total}}} = 1 - \frac{\sum e_i^2}{\sum (Y_i - \bar{Y})^2}$$

- ① $R^2 \in (0, 1)$
- ② R^2 is an increasing function of the number of variables
- ③ Higher value of R^2 implies a better model.

Adjusted R^2 :

gives us an appropriate time to stop.

$$R_{\text{adj}}^2 = 1 - \frac{\sum e_i^2 / (n - k - 1)}{\sum (Y_i - \bar{Y})^2 / (n - 1)} < R^2$$

★ If $L^T y$ is an LVE of $p^T \beta$, then $L^T P_X y$ is the BLUE of $p^T \beta$

★★ The if & only if condition for $p^T \beta$ to be Estimable is $p \in C(X^T)$

Proof :- $E(L^T y) = L^T p^T \beta \quad \forall \beta \in \mathbb{R}^{k+1}$
 $\Rightarrow L^T X \beta = p^T \beta \quad \Rightarrow L^T X = p^T$

So, $p^T = X^T L \Rightarrow p \in C(X^T)$

★★★ Show that $L^T y$ is an LZF iff $X^T L = 0$
 By definition: $E(L^T y) = 0 \quad \forall \beta \in \mathbb{R}^{k+1}$
 $\Rightarrow L^T X \beta = 0 \Rightarrow L^T X = 0^T \quad [0 = 0^T \beta]$

$\therefore L \in (C(X))^{\perp}$ [Anything which is a linear combination of X , if we take dot product with L^T , we get 0]

$\Rightarrow L \in C(I_n - P_X)$; If this is true, there will exist some m such that $L = (I_n - P_X)m$

Now we can see that :- $L^T X = [(I_n - P_X)m]^T X = m^T (I_n - P_X) X$
 $= \sum m^T L = 0$

Thus, every L.Z.F. will be of form $m^T (I_n - P_X) X$

Let $L^T y$ be a LVE of $p^T \beta \Rightarrow E(L^T y) = p^T \beta$; Then the BLUE of $p^T \beta$ will be $L^T P_X y$

Proof :- ① Note that $L^T y = L^T I_n y = L^T (P_X + I_n - P_X) y$
 $= L^T (I_n - P_X) y + L^T P_X y$

② $E(L^T P_X y) = L^T P_X X \beta = (L^T X) \beta = p^T \beta$; It is a LVE
 ③ $E(L^T (I_n - P_X) y) = L^T (I_n - P_X) X \beta = 0$; It is a LZF

Now, Cov $(L^T P_X y, L^T (I_n - P_X) y) = L^T P_X (\sigma^2 I_n) (I_n - P_X) L$
 $= \sigma^2 L^T P_X (I_n - P_X) L = 0$

Thus, $L^T P_X y$ is the BLUE of $p^T \beta$

Note :- BLUE is always Unique.

Proof :- Let, if possible, $L_1^T y$ & $L_2^T y$ both be the BLUE's of $p^T \beta$

Now, $l_1^T y = l_2^T y + (l_1 - l_2)^T y$
 Taking E on both sides $\rightarrow$

$E(l_1^T y) = p^T \beta$ Also true for $l_2^T y$
 $\therefore E((l_1 - l_2)^T y) = 0$; Also $Cov(l_1^T y, (l_1 - l_2)^T y) = 0$

So, $Var(l_1^T y) = Var(l_2^T y) + Var((l_1 - l_2)^T y)$

Case - 1 :- $Var(l_1^T y) > Var(l_2^T y)$ Contradiction

Case - 2 :- $Var((l_1 - l_2)^T y) = E((l_1 - l_2)^T y) = 0$

$\therefore P[(l_1 - l_2)^T y = 0] = 1 \Rightarrow P[l_1 = l_2] = 1$